

Questions			Answers	Marks
1	(a)	(i)	The gas molecules move randomly (and rapidly)	B1
		(ii)	The <u>volume</u> of their <u>molecules</u> is negligible compared to the volume of the containing vessel.	B1
	(b)	(i)	The gas molecules move in three dimensions (or x- , y- and z-direction.)	B1
		(ii)	Using $p = \frac{1}{3}\rho \langle c^2 \rangle$, we get $101\,000 = \frac{1}{3} \times 1.25 \times \sqrt{v}$	M1
			$v = 492 \text{ m s}^{-1}$	A1
		(iii)	Kinetic energy of ideal gas $\propto T$, and Kinetic energy of ideal gas $\propto v_{rms}^2$ so $v_{rms}^2 \propto T$	M1
			$\frac{v}{492.34} = \sqrt{\frac{273.15 + 127}{273.15}}$ $v = 596 \text{ m s}^{-1}$ award M1 mark if temperature not given in thermodynamic temperature, but A0	A1